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17 September 1981
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21 March 1974

MILITARY SPECIFICATION

RECORDER, CODE MATRIX BLOCK LA-434( )

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of recording head assembly designated Recorder, Code Matrix Block, LA-434( ).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6720

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SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Signal Equipment.
MIL-N-18307	- Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Group Support Equipment.
MIL-E-55585	- Electronics Equipment and Parts, Packaging of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electronmagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-781	- Reliability Tests: Exponential Distributions.
MIL-STD-810	- Environmental Test Methods.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample units for first article inspection and approval (see 4.3 and 6.2e).

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3.2 Parts, material and processes. In addition to the requirements of this specification, MIL-P-11268 shall be the prime specification for the selection of electronic parts, materials and processes including microelectronics.

3.2.1 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.3 Operational requirements.

3.3.1 Matrix pattern. The data matrix produced by the pattern generator on the cathode ray tube of the equipment, when tested in accordance with 4.7, shall consist of three columns of binary, or alpha numerical, coded decimal characters as shown in figure 1 (see 4.7.1.1).

3.3.2 Matrix densities. The matrix densities of the pattern produced by the pattern generator on the cathode ray tube of the equipment when tested in accordance with 4.7 shall be measured as specified in 4.7.1.2 and shall conform to figures 2 and 3.

3.3.3 Matrix geometry. The matrix geometry shall meet the following requirements when measured as specified in 4.7.1.3 after testing in accordance with 4.7:

a. No index dot center of line A-C shall depart more than ± 0.009 inch (.229mm) from a best fit reference line defined by the index dot center lines parallel to reference line A of figure 4.

b. No index dot center of line A-B shall depart more than ± 0.012 inch (.305mm) from a best fit reference line defined by the index dot center lines parallel to reference line B of figure 4.

c. The A-C and A-B dot rows shall be perpendicular within ±3°.

d. Index row parallel to edge of film format within ±3°.

e. Matrix geometry shall not deviate from boundary dimensions as stated in figure 4.

3.3.4 Matrix dot size. The diameter of each dot in the photographic image of the recorder code matrix block display, as recorded on the film negative (see 4.7.2) of the associated sensors, and with a microdensitometer having a 1-mil scanning aperture, shall be .008 ± .002 inches (.203 ± .051mm) at the points where the density is 50 percent of the difference between the peak dot density and the background density. The background density is defined as that density which is measured by a microdensitometer having a 1-mil scanning aperture, at a point on the photographic film within the data matrix which contains no dot centers within 0.015 inches (.381mm) distance from that point.

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3.3.5 Matrix dot contrast. The dots of the data matrix when presented on the face of a recorder code matrix block for which the exposure controls of the aircraft camera control system have been properly adjusted and when directed by the recording heads self-contained fiber optic transporter upon a photographic film (see 4.7.2) having an aerial exposure index (AEI) rating of 6 thru 200 and developed to any specified gamma within the range recommended according to film type, shall have a minimum density difference of 0.2 between the peak dot density and the inter-dot density (see figure 2).

3.3.6 Matrix isolated dot characteristics. An isolated dot shall be defined as a dot whose center is at least 0.023 inch (.58mm) away from all other dot centers. An isolated dot shall meet the shape and contrast requirements specified below (see 3.3.6.1).

3.3.6.1 Matrix isolated dot shape. The characteristic shape of an isolated dot is based on scanning through the center of the dot with a 1-mil spot size recording microdensitometer. The record shown in this manner shall fall within the envelope of acceptability shown in figure 2 except that there may be one spike, extending downward into area A, resulting from a single blemish no larger than .0015 in. (.098mm) in diameter. The required shape as shown in figure 2 is based on absolute distance scale and a relative density scale with 100 percent being defined as the peak dot density and zero as the background density.

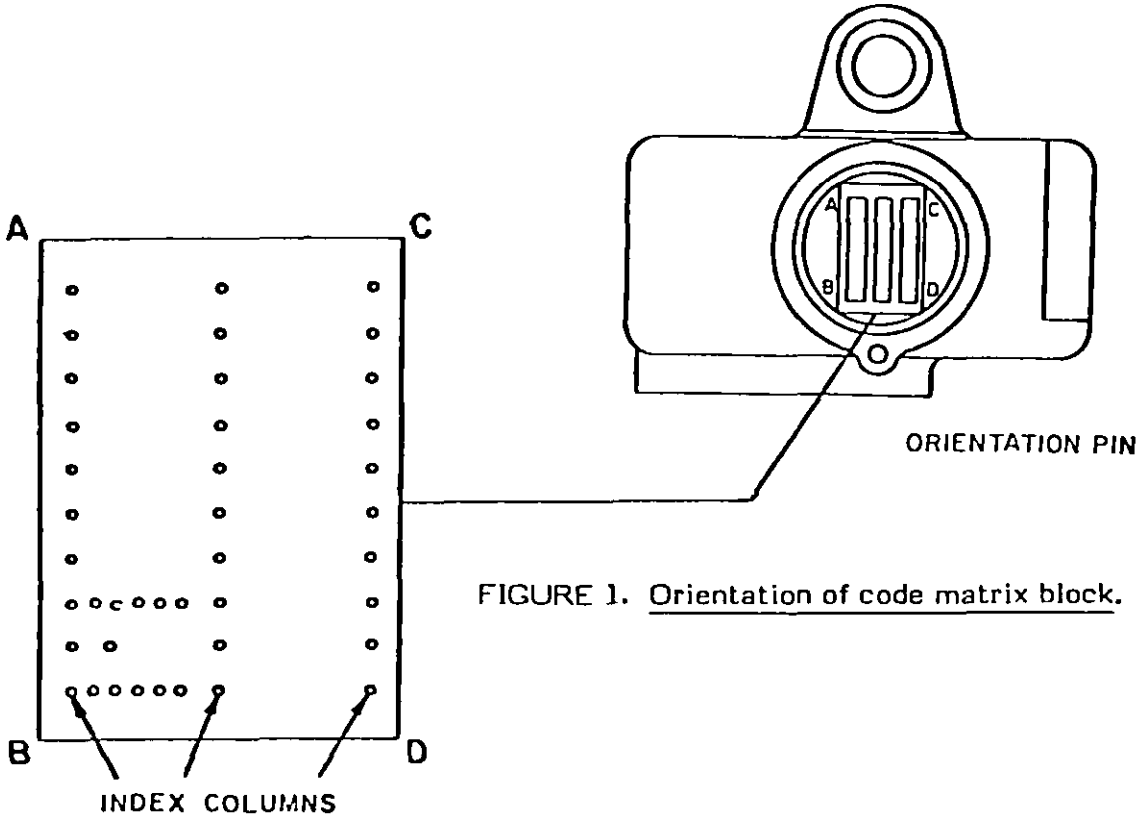


FIGURE 1. Orientation of code matrix block.

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PEAK DOT DENSITY IS 100%

BACKGROUND DENSITY OF 0%

AREA OF ACCEPTABILITY OF MICRODENSITOMETER TRACING

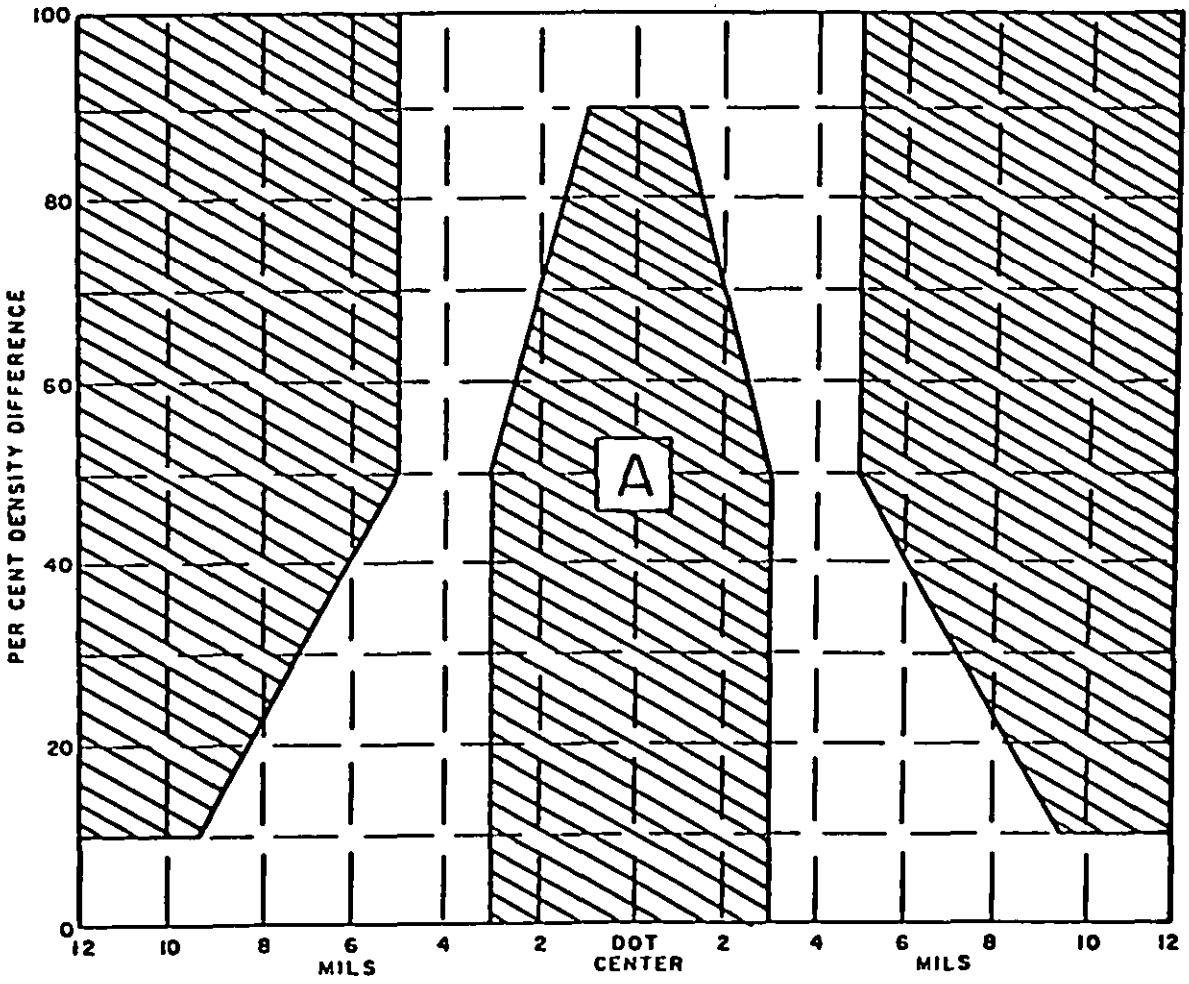


FIGURE 2. Isolated dot shape requirements.

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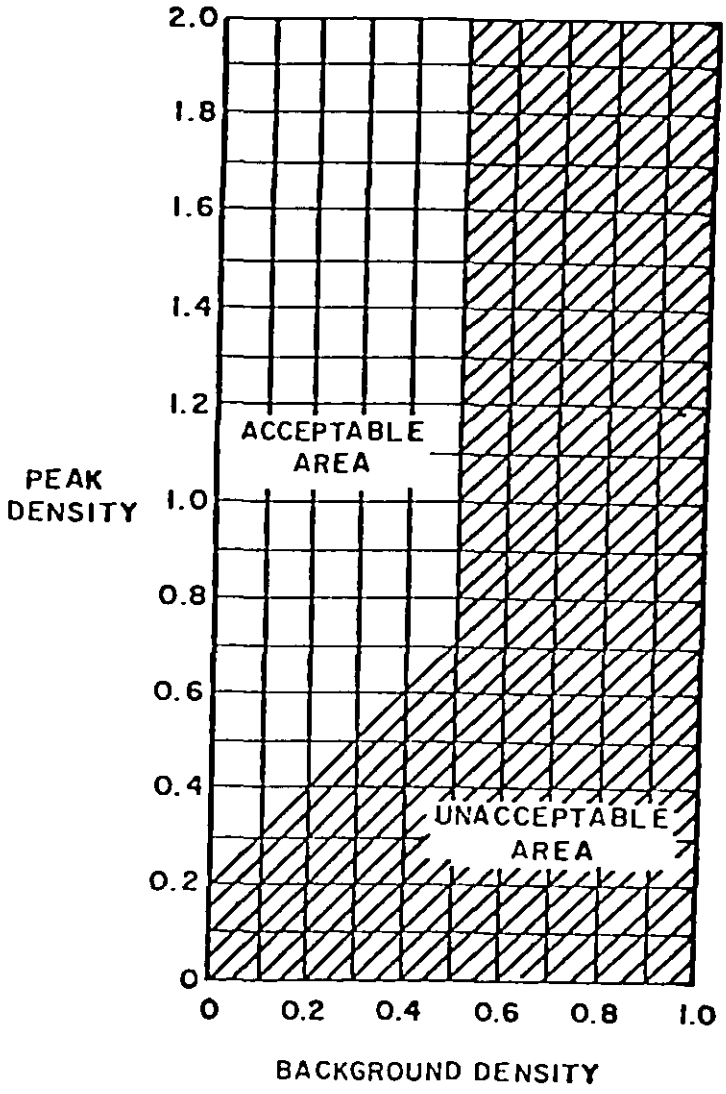


FIGURE 3. Requirements for the relationship of peak density to background density for an isolated dot developed to a gamma of 1.2.

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NOTES:

1. No index dot center shall depart horizontally more than +.003 inch (.076mm) from a line drawn parallel to reference line "B" through the dot center of the adjacent index dot. In addition, the maximum horizontal departure of any index dot center from a line drawn parallel to reference line "B" through the index dot center of row 32 shall not exceed ±.012 inch (.305mm).
2. Within any given digit, no bit dot center shall depart more than +.003 inch (.076mm) from a line drawn through the center of its index dot parallel to reference line "A."
3. Reference lines "A" and "B" are best fit straight lines as shown in figure 4.
4. With the matrix oriented as shown, "BCD" digits shall always be presented with the most significant decimal digit at the bottom of the block and the least significant decimal digit at the top of the block.
5. All of the dots shown in this figure shall appear as visible dots on a photographic range.

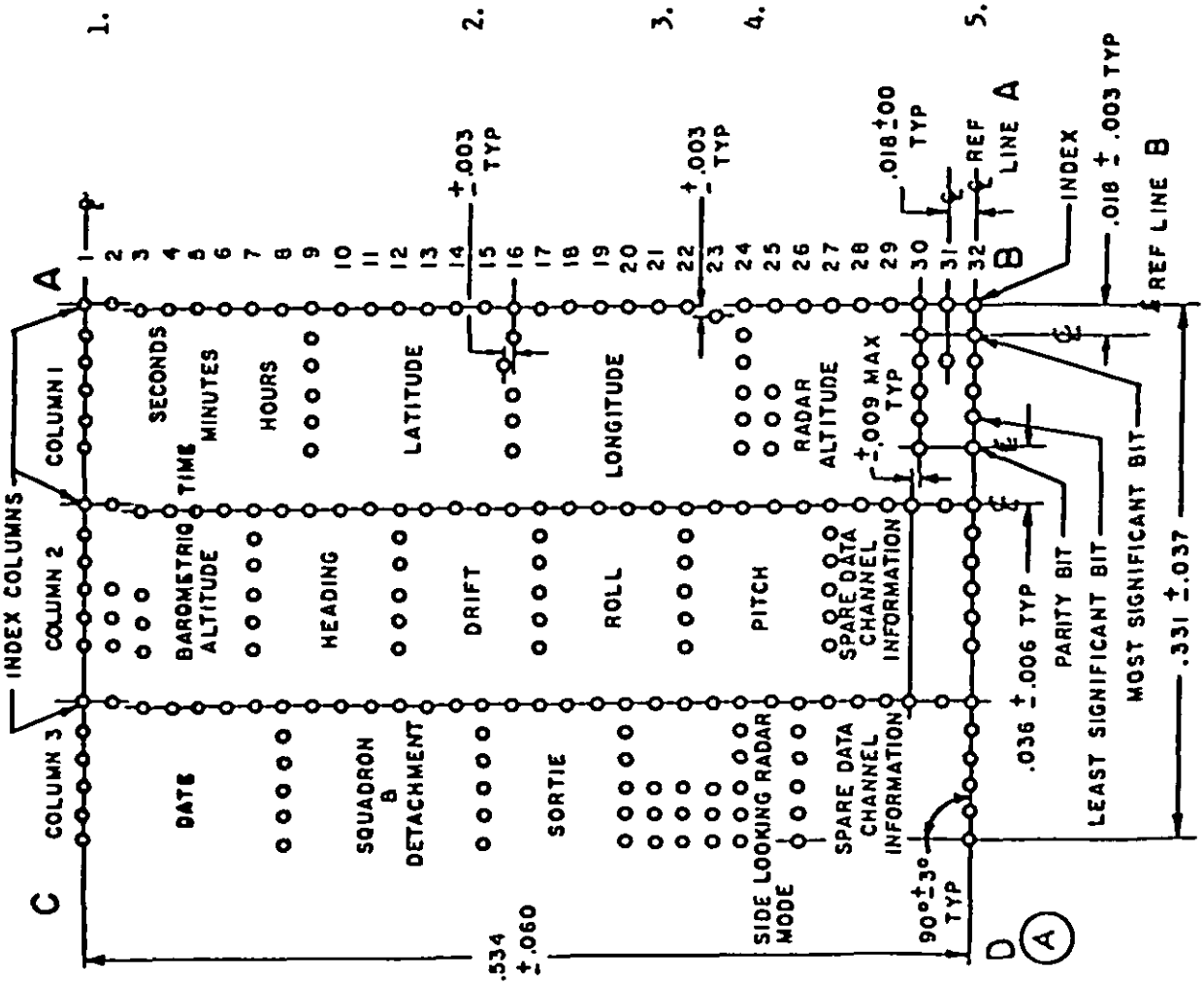


FIGURE 4. Data format.

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3.3.6.2 Matrix isolated dot contrast. Isolated dots of a data matrix, exposed on film (see 4.7) and developed as specified in 3.3.6.1 above, shall meet the requirements of figure 3 which shows the area of acceptability for isolated dots.

3.4 Attitude. The equipment shall be capable of meeting the operational requirements of 3.3 in a horizontal to vertical attitude (see 4.10).

3.5 Reliability. The Recorder, Code Matrix Block LA-434( ), shall have a mean-time-betweeen-failure (MTBF) of 1000 hours when operated under the following conditions:

a. Temperature: -18°C (0°F) to +49°C (120°F).

b. Vibration: 2.2g ± 10 percent at 29 Hz.

c. Voltage cycling: None.

d. On-off cycling: 3 hours on, 1 hour off.

e. Utilization: 107 pulses per hour of equipment on time consisting of 64 pulses at 6 pulses per second, 32 pulses at 2 pulses per second, and 11 pulses at 1 pulse per second.

3.6 Electromagnetic compatibility. The equipment shall meet the requirements specified in table A-VI-MIL-I-6181D of MIL-STD-461 (see 4.9). The generation of electromagnetic interference by the equipment shall be within the limits therein with the following exceptions:

a. The broadband conducted limits shall be within +15db of the limits stated.

b. The high frequency broadband radiated limits shall be within +25db at 25MHz decreasing linearly to the limit at 300 MHz specified in the standard.

3.7 Fungus resistance. Materials not nutrients for fungus growth shall be used to the maximum extent possible. All other materials shall be suitably finished or treated (see MIL-P-11268) to resist the growth of fungi. Anti-fungus treatments that are deleterious to the photographic emulsions shall not be used. The equipment shall show no evidence of viable fungus or corrosion after the fungus test (see 4.18).

3.8 Service conditions. The equipment shall meet the following service conditions:

3.8.1 Temperature.

3.8.1.1 Low. Before, during and after the test specified in 4.11.1, the equipment shall meet the requirement of 3.3 at -18°C (0°F) (see 4.7).

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3.8.1.2 High. Before, during and after the test specified in 4.11.2, the equipment shall meet the requirement of 3.3 at 49°C (120°F) (see 4.7).

3.8.2 Humidity. Before, during and after the test specified in 4.12, the equipment shall meet the requirement of 3.3 (see 4.7).

3.8.3 Altitude. Before, during and after the test specified in 4.13, the equipment shall meet the requirement of 3.3 (see 4.7).

3.8.4 Vibration. Before, during and after the test specified in 4.14, the equipment shall meet the requirement of 3.3 (see 4.7).

3.8.5 Shock. Before and after the test specified in 4.15, the equipment shall meet the requirement of 3.3 (see 4.7). Any physical damage shall be minor only.

3.8.6 Acceleration. Before, during and after the test specified in 4.16, the equipment shall meet the requirement of 3.3 (see 4.7).

3.8.7 Explosive atmosphere. The equipment shall not cause ignition of any surrounding air-vapor mixture when operating in an explosive atmosphere (see 4.7).

3.9 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as covers, screws, etc., and necessary touch-up after mounting. The final paint film on type 1 surfaces shall be semi-gloss enamel matching a color chip provided by the procuring agency (see 6.3). Protective treatment shall not be used on machine finished surfaces such as mating surfaces, etc., if it impairs the function of the part. The use of protective coating that will crack, chip or scale with age or extremes of climate and environmental conditions shall be avoided.

3.9.1 Interior. Surfaces which are exposed to the optical path shall be finished in a durable, dead matte black. Finish may be omitted for electrical wiring.

3.10 Identification of product. The equipment shall be marked for identification in accordance with MIL-M-13231. The configuration for the nameplate shall be in accordance with figure 4 of MIL-N-18307, and as specified in the contract.

3.10.1 Serial numbers. A serial number shall be assigned by the contractor to each recorder. These numbers shall not be duplicated in the course of any one contract.

3.11 Tube, outside diameter. The insertable tubular portion of the Recorder, Code Matrix Block LA-434( ) (see 4.19) shall conform to the following dimensions:

Length: 3.25 (+0.015 or -0.003) inches (82.55 (+0.381 or -0.066) mm)

Diameter: 0.985 (+0 or -0.002) inches (25.02 (+0 or -0.051) mm)

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3.12 Burn-in. Burn-in shall be performed in accordance with 4.23.

3.13 Workmanship. The equipment shall be manufactured and assembled in accordance with MIL-STD-454 requirement 9 (see 4.20).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.2); (does not include packaging).
b. Inspection covered by subsidiary documents (see 4.4).
c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before packaging (see 4.5).
(2) Quality conformance inspection for packaging (see 4.21).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish first article units of the Recorder, Code Matrix Block LA-434( ) (see 6.2e).

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

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TABLE I. Order of first article inspection.

Inspection	Requirement paragraph	Test method paragraph	Unit 1	Unit 2	Units 3 - 5
Finish	3.9 & 3.9.1	4.4	Inspection to be performed on all units		
Wiring	3.2	4.4			
Visual & mechanical	3.11 & 3.13	4.19 & 4.20	Inspection to be performed on all units		
Operation	3.3	4.7 thru 4.7.2			
Attitude	3.4	4.10	Inspection to be performed on all units		
Electromagnetic compatibility	3.6	4.9	1		
Temperature (low)	3.8.1.1	4.11.1		1	
Temperature (high)	3.8.1.2	4.11.2		2	
Altitude	3.8.3	4.13		3	
Explosion proof	3.8.7	4.17		4	
Vibration	3.8.4	4.14		5	
Humidity	3.8.2	4.12		6	
Acceleration	3.8.6	4.16		7	
Shock	3.8.5	4.15		8	
Fungus 1/	3.7	4.18		9	
Reliability	3.5	4.8			1

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract:

Item	Where required
Finish	3.9, 3.9.1
Parts, materials and processes	3.2

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4.5 Quality and conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels, and AQLs indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective)	
			Major	Minor
Visual and mechanical Operation	3.11, 3.13 3.3	4.19, 4.20 4.7 thru 4.7.2	1 1	4 1/

1/ All defects are to be considered major.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The group B AQL shall be as shown in table III and the inspection level shall be S-4.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (% defective)
Attitude	3.4	4.10	4

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Three units, selected at random in accordance with table V below shall be subjected to the tests of table IV as follows: two for environmental, and one for EMI. The environmental tests shall be divided between two units as follows: Unit 1: Altitude, explosion proof, vibration, acceleration, shock, fungus; Unit 2: Low temperature, high temperature, humidity.

4.5.3.2 Order of inspection within group C. Group C inspection shall be performed in accordance with table IV.

4.5.3.3 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.4 Reinspection of conforming group B and group C. Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup I		
Temperature (low)	3.8.1.1	4.11.1
Temperature (high)	3.8.1.2	4.11.2
Altitude	3.8.3	4.13
Explosion proof	3.8.7	4.17
Vibration	3.8.4	4.14
Humidity	3.8.2	4.12
Acceleration	3.8.6	4.16
Shock	3.8.5	4.15
Fungus 1/	3.7	4.18
Subgroup II		
Electromagnetic compatibility	3.6	4.9

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after inspection before proceeding with subsequent inspection.

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TABLE V. Sampling for group C inspection.

Number of units on order	Quantity requiring inspection
Up to 100	3 units from first 50 produced, 3 units from last 50 produced.
101 - 500	3 units from first 100 produced, 3 units per each 100 units thereafter.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table VI and shall be performed on units from lots which have been subjected to and met group A and group B inspection and the electrical inspection listed under subgroup II of group C inspection in table IV.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test method paragraph
Reliability	3.5	4.8

4.5.5.1 Sampling for group D inspection. Samples shall be selected at random and the sample size for each reliability test shall be not less than three units nor more than five units.

4.5.5.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract.

4.5.6 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be accepted on the contract provided all visible and known damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Standard conditions. Unless otherwise specified, the following conditions shall be used as a basis for establishing normal performance limits for all tests to be conducted:

Temperature:	77° ± 18°F (25° ± 10°C)
Relative humidity:	Room ambient up to 90%
Vibration:	None
Barometric pressure:	28 to 32 inches (71.1 to 81.3cm) of mercury

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4.7 Operational test. Set the writing time of the pattern generator (figure 5) as shown in table VII, load the camera or camera simulator with EK-8401 (Plus-X) film. Carefully install the recorder in the camera or camera simulator. Set the command/continuous switch to "continuous." Apply power to the pattern generator and observe the pattern on the tube. Return the command/continuous switch to "command." If camera is used, turn on power to the camera operating fixture and vacuum source. Trip the camera several times and make certain a display is flashed on the recorder once per frame. If camera simulator is used manually, expose 10-20 frames using a manual command switch. Operate the command switch once for each display. If camera is used, turn on light box and operate the camera for 10-20 frames in the pulse mode. Turn off the light box and advance a minimum of 10 additional frames (see 3.3 thru 3.3.6.2).

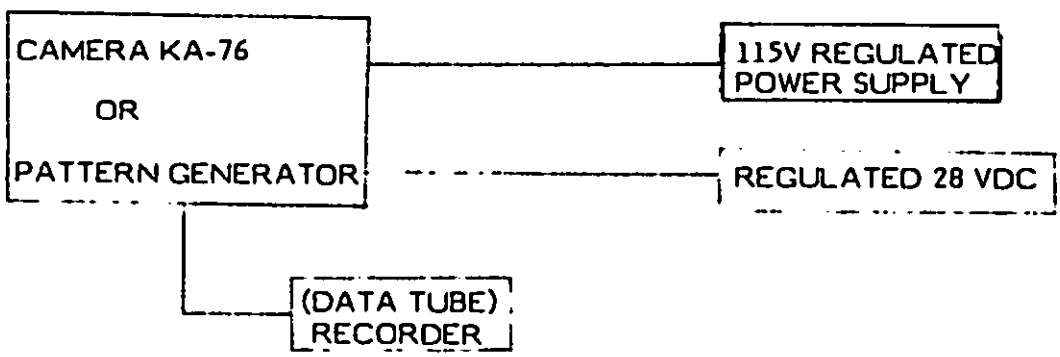


FIGURE 5. Recorder test hookup.

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TABLE VII. Writing time of pattern generator.

Pattern generator setting (see 4.7)	Development chemistry	Gamma
5 sec. (position 4)	D-76	1.2
5 sec. (position 4)	MS-641 (Versimat)	1.2
20 sec. (position 6)	A Chem (Versimat)	1.2
20 sec. (position 6)	D-19	1.2

4.7.1 Evaluation for satisfactory operation.

4.7.1.1 Matrix pattern. The index corners and column shall be similar to figure 1 (see 3.3.1).

4.7.1.2 Matrix densities. Measure dot densities of a single row plus an isolated dot on a microdensitometer with a .001 inch (.025mm) aperture, using the film exposed in operational test (see 4.7). The dot densities shall meet the requirements of 3.3.2.

4.7.1.3 Matrix geometry. Using the optional comparator and a mask suitably inscribed with the digital data matrix boundaries and reference lines (figure 4) perform measurements on the negatives obtained from the operational test in 4.7 to assure that the requirements of 3.3.3 are met.

4.7.2 Matrix dot test. The dot size, contrast and isolated dot characteristics including dot shape, shall be inspected after being submitted to the test in 4.7 to determine compliance with the requirements of 3.3.4 through 3.3.6.2.

4.8 Reliability.

4.8.1 Reliability test. The contractor shall perform a reliability test for each quantity of 100 units produced or fraction thereof. The test shall be performed in accordance with test plan IV of MIL-STD-781. The minimum test time per equipment shall be in accordance with 4.2.2.4 of MIL-STD-781.

4.8.2 Test conditions. All testing shall be conducted under the environment of test level E of MIL-STD-781 except as follows:

a. Temperature cycling: -18°C (0°F) to +49°C (120°F).

b. On-off cycling: 3 hours "on," 1 hour "off."

c. Vibration: 2.2g ± 10 percent at 29 Hz for 10 minutes of every hour that the equipment is "on."

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4.8.3 Test procedure. During each hour that equipment is operated, the recorders shall be pulsed 107 times. The 107 pulses shall consist of 64 pulses at 6 pulses per second, 32 pulses at 2 pulses per second, and 11 pulses at 1 pulse per second. Each day all voltages applied to the recorders shall be checked. Orientation of the dot pattern on the recorders shall be checked daily. Dot pattern presence shall be continuously monitored. At approximately every 200 hours of total operating time, a live film test shall be performed on all 3 operating recorders. After completion of the required total hours for acceptance of the test, each LA-434( ) shall pass group A inspection requirements.

4.8.4 Failure definition. A failure is determined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractors approved test procedures. Paragraphs 5.5 through 5.10 of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with 5.4.8 of MIL-STD-781.

4.8.5 Rework and retest provisions. Once a reject decision has been reached no rerun can be made of that test on the same or a new sample from the lot unless positive action consisting of design change, modification or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.9 Electromagnetic compatibility test (see 3.6). Test methods and procedures shall be in accordance with MIL-STD-462, with the following exceptions:

a. Radiated interference test shall be performed from .014 to 1000 MHz.

b. Radiated susceptibility test shall be performed from .015 to 400 MHz.

4.10 Attitude tests. The recorder shall be operated in accordance with 4.7 in all modes in at least six different positions from the horizontal to vertical.

4.11 Temperature tests.

4.11.1 Temperature (low). The equipment shall be subjected to the test of method 502, procedure I, of MIL-STD-810. The storage temperature (step 2) shall be -57°C (-70°F) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating (step 4) shall be -18°C (0°F).

4.11.2 Temperature (high). The equipment shall be subjected to the test of method 501, procedure II, of MIL-STD-810 with the exception that the temperature for step 4 shall be 71°C (160°F). The operating temperature (step 7) shall be 49°C (120°F).

4.12 Humidity test. The equipment shall be subjected to the test of method 507, procedure II, of MIL-STD-810. During the test the lens cap shall be installed. In step 6, the operation of the equipment shall be checked during the last 5 hours of the 2nd and 5th cycles.

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4.13 Altitude test. The equipment shall be subjected to the test of method 500, procedure I, of MIL-STD-810, with the exception that for test item operation in step 3 the chamber conditions shall be set at -18°C (0°F) and 8.3 in. (2.1cm) of Hg (equivalent to 30,000 ft or 9144m, above sea level). The test item shall be stabilized at these conditions prior to operation.

4.14 Vibration test. The equipment shall be subjected to the test of method 514.1, procedure I, of MIL-STD-810B. The test shall be in accordance with table 514.1 - II and figure 514.1-2, part 1 (curve C).

4.15 Shock test. The equipment shall be subjected to the test of method 516.1, procedure I and III, of MIL-STD-810. The shock pulses shall be in accordance with figure 516.1-2 for flight vehicle equipment.

4.16 Acceleration test. The equipment shall be subjected to the test of method 513.1, procedures I and II of MIL-STD-810. The test level shall be 10g for procedure I and 5g for procedure II.

4.17 Explosive atmosphere test. The equipment shall be subjected to the test of method 511, procedure I, of MIL-STD-810.

4.18 Fungus test. The equipment shall be subjected to the test of method 508, procedure I of MIL-STD-810. Areas of the test item which are sealed by gasket or other non-hermetic type seal shall be opened during test exposure and all internal as well as external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required.

4.19 Inspection for dimensional integrity. The insertable dimensions of the recorder shall be gaged or measured to determine conformance to the physical sizes specified in 3.11. When a listed dimension does not conform to the above, it shall be considered a major defect.

4.20 Visual and mechanical. Equipment shall be examined for the defects listed in MIL-STD-252 in addition to those listed in table VIII.

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TABLE VIII. Classification of visual and mechanical defects.

Classification	Defect
Major	- Electrical functions inoperable. - Equipment does not secure in place. - Connecting cables insufficient length. - Missing, inoperative, or incorrect part which could cause equipment to be inoperative or unsafe in service.
Minor	- Safety or warning marks missing, illegible, or incorrect. - Equipment identification missing, illegible, or incorrect. - Slight damage or dents in case which will not prevent fit.

4.21 Quality conformance inspection of packaging. Quality conformance inspection of packaging requirements shall be in accordance with the group A, group B and when required (see 6.2f), the group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

4.22 Rough handling test (packaging). When rough handling test in accordance with specification MIL-P-116 is required by the contract (see 6.2f), the operational test (see 4.7) shall be conducted to determine freedom from operational malfunction caused by the rough handling.

4.23 Burn-in.

4.23.1 Sequence of testing. Before subjecting equipment to this procedure, the contractor must have available objective evidence of compliance to group A requirements.

4.23.2 Test conditions. Every equipment shall be subjected to a 24 hour burn-in period. The burn-in shall be conducted under the conditions for reliability testing. The failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF.

4.23.3 Compliances. Units completing burn-in shall be submitted for group A, B and C inspections in that order without benefit of repair, cleaning, realignment or other changes, unless otherwise specified herein. If after successful completion of burn-in, the equipment fails either group A or B tests, the equipment will be reworked and resubjected to burn-in.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-NU-X-B7-E-ED-A.

5.1.2 Level B. 1Ø-1-1-ØØ-NU-X-B7-E-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The Recorder, Code Matrix Block LA-434( ) is used to display binary decimal or alpha numerical information provided by the data annotation system in the aircraft upon receipt of a command signal from the aircraft camera control system to the camera, still picture KA-76( ), to be recorded on the emulsion through the back of the film.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see section 5).

c. Marking and shipping of samples.

d. Place of final inspection.

e. Number of first articles to be submitted for approval (see 3.1).

NOTE: Five sample units of each item cited in section 1 are normally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the other sample units (see 4.3.1).

f. When rough handling test is required (see 4.22).

g. Nameplate data.

6.3 Color. The color chip furnished by the procuring agency will be color chip no. X24087, and may be obtained upon request to the contracting officer.

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6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example, LA-434W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army - ER

Preparing activity:
Army - ER
(Project 6720-A312)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-49016A(ER) RECORDER, CODE MATRIX BLOCK LA-434( )

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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